

1	2	3	4	5	6	7	8
SAM CPU replacement board, draft 0.2, drawn from docs/ARCHITECTURE.md. Licence: CERN–OHL–S v2. A Raspberry Pi 4 on the 40–pin header J21 runs PinMAME + ppuc (HDMI stays on the Pi). The RP2354B drives the original IO power driver board on J9 and scans the switches. Connector names and pinouts follow the original CPU/Sound board 520–5246–00. Each connector sits on the same page as the circuit that drives or reads it. Signals between pages use global labels; the numbers in brackets next to a label list the other pages it goes to.							
A							
B	Power Power: +5 V mux (external / J11), 3.3 V, 4.5 V, supply monitors File: power.kicad_sch		MCU and Pi RP2354B SAM IO controller and Raspberry Pi 4 header File: mcu.kicad_sch		IO bus J9: bus to the IO power driver board (IO board J1) File: io_bus.kicad_sch		
C	Switch columns Switch matrix columns: 74HC595, strobe drivers, J1 File: sw_columns.kicad_sch		Switch rows 1–8 Switch matrix returns 1–8: J6, LM339 front end, 74HC165 File: sw_rows_1.kicad_sch		Switch rows 9–16 Switch matrix returns 9–16: J12, LM339 front end, 74HC165 File: sw_rows_9.kicad_sch		
D	Dedicated switches 1–16 Dedicated switches D1–D16 (J2, J3) and coin door memory protect File: sw_dedicated_1.kicad_sch		Dedicated switches 17–24 Dedicated switches D17–D24 (J13) and DIP switches File: sw_dedicated_2.kicad_sch		Display and GI J5 original DMD driver and J18 GI dimmer header File: display_gi.kicad_sch		
E	Audio Audio: Pi I2S → PCM5102A → 2 x TDA2030A on +−12 V → J10 File: audio.kicad_sch						
F	1	2	3	4	5	6	8

Stern SAM CPU PPUC (CERN–OHL–S v2)

Sheet: /
File: sam_cpu.kicad_sch

Title: SAM CPU replacement board

Size: A3

Date: 2026–10–06

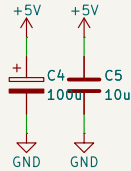
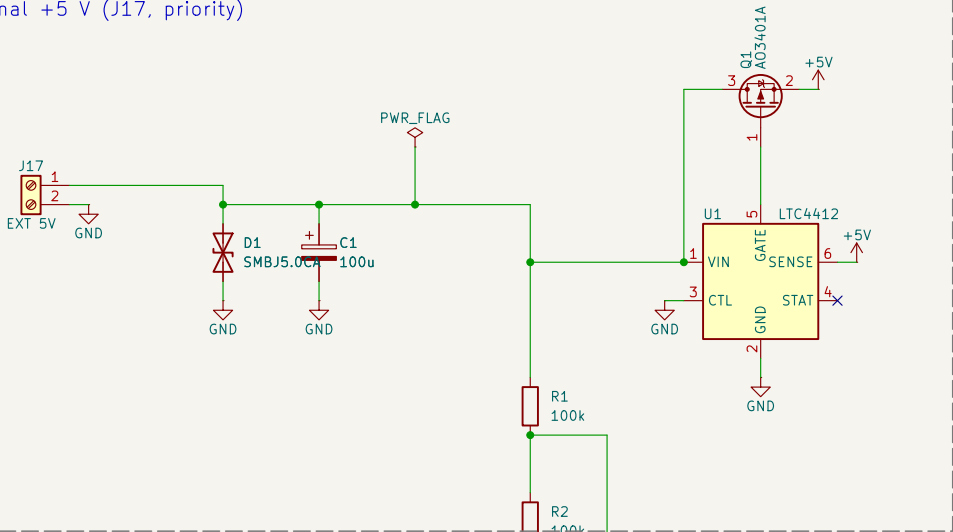
Rev: 0.2

KiCad E.D.A. 10.0.6

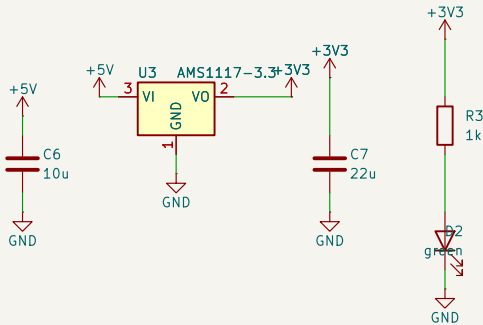
Id: 1/11

J11 keeps the original CPU board pinout (IO board J16 harness): +5 V, +−12 V and ground.
External +5 V on J17 has priority: U2 turns the J11 path off whenever +5V_EXT is present (EXT_PRESENT high).
LTC4412 + A03401A ideal diodes stand in for the TPS2121 of ARCHITECTURE.md 9.3 (library part, about 4 A per path).
JP1 (bridged by default) disconnects the J11 +5 V completely: leaving J17 unplugged forces J11.

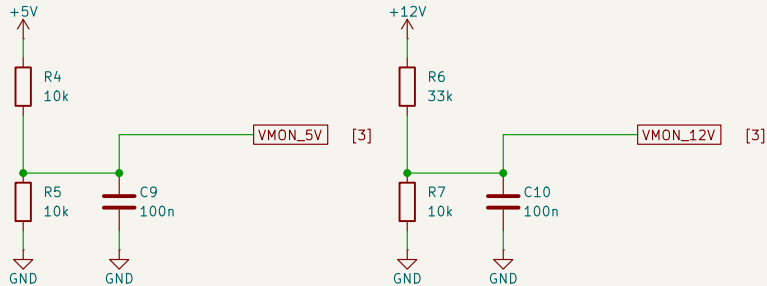
External +5 V (J17, priority)



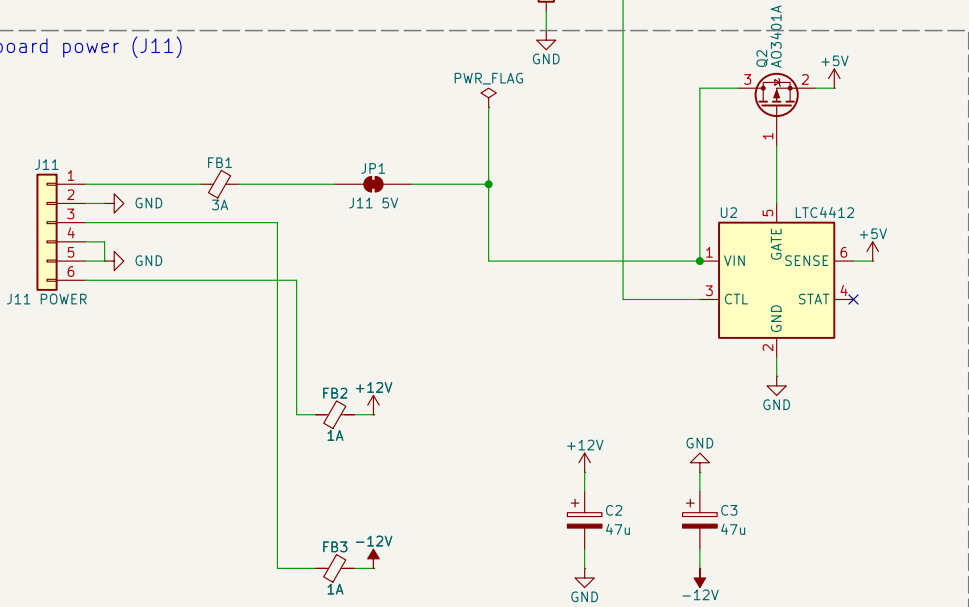
3.3 V



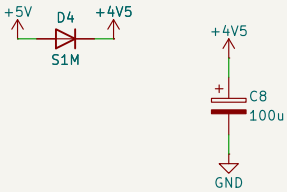
Supply monitors (RP2354B ADC)



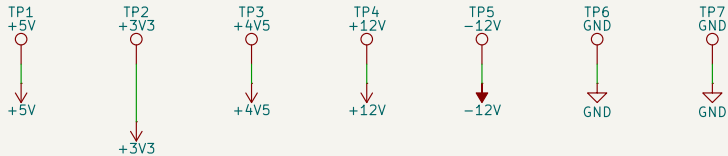
IO board power (J11)



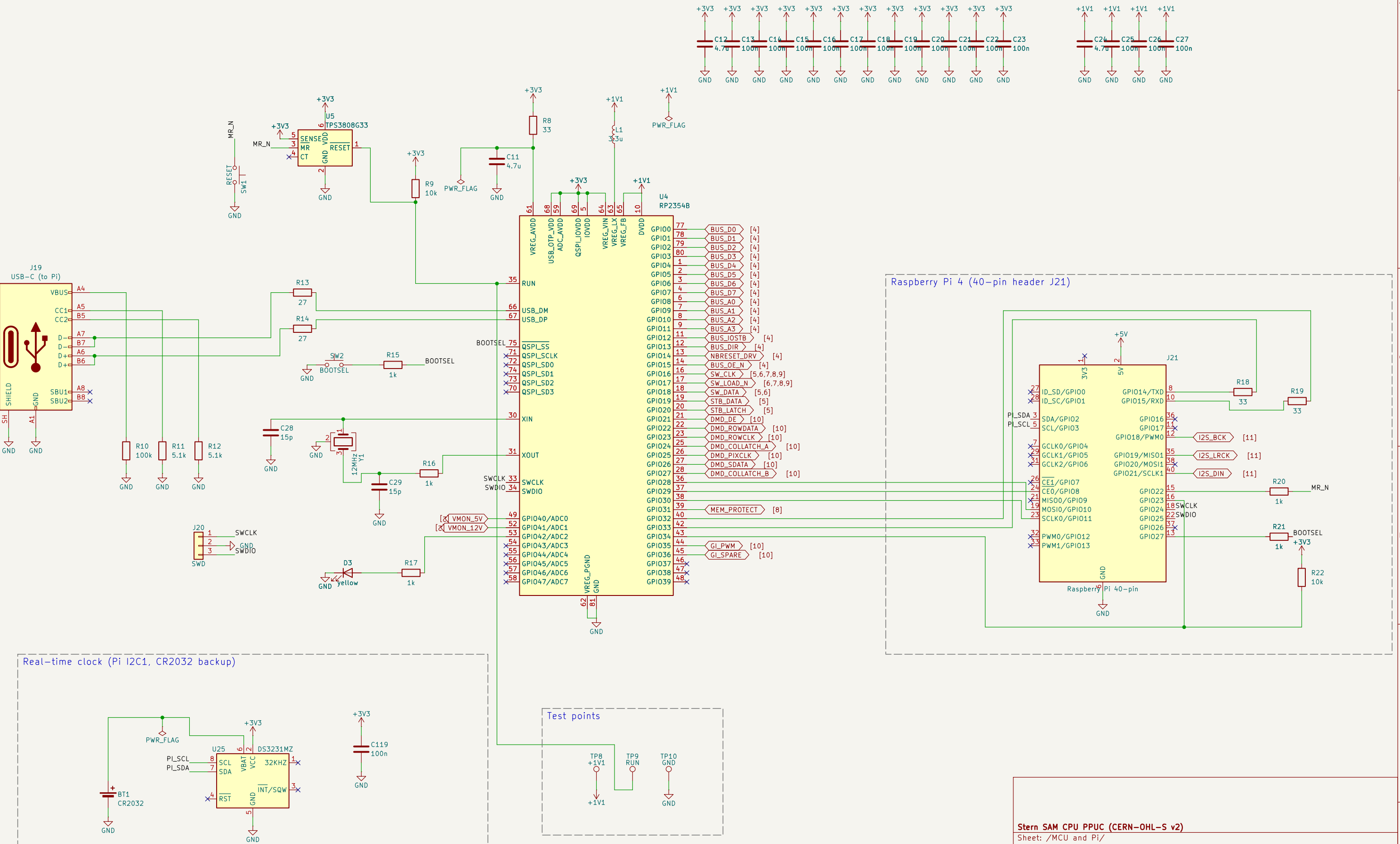
4.5 V for the switch matrix



Test points



GPIO map: ARCHITECTURE.md 9.2. PIO0 bus on GPIO0–15, PIO1 switch chain on GPIO16–20, PIO2 DMD on GPIO21–27, GPIO31 coin door memory protect.
Core supply from the internal switching regulator: VREG_LX -> L1 -> DVDD (1.1 V); VREG_AVDD through 33 R / 4.7 uF. Check against the RP2350 hardware guide.
Raspberry Pi 4 on the 40-pin header J21 (HDMI stays on the Pi). The Pi is powered from the board +5 V; never plug the Pi USB-C supply at the same time.
U25 DS3231MZ real-time clock on the Pi I2C1 bus (Linux: dtoverlay=i2c-rtc.ds3231), backed by a CR2032 in BT1.



ARCHITECTURE.md 3.1–3.2. The doc calls this connector J1 (its name on the IO board); on the CPU board it is J9. Both buffers are off at power-up (BUS_OE_N pulled high) and NBRESET is held low until the firmware releases it. 33 R series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.

A

B

C

D

E

F

A

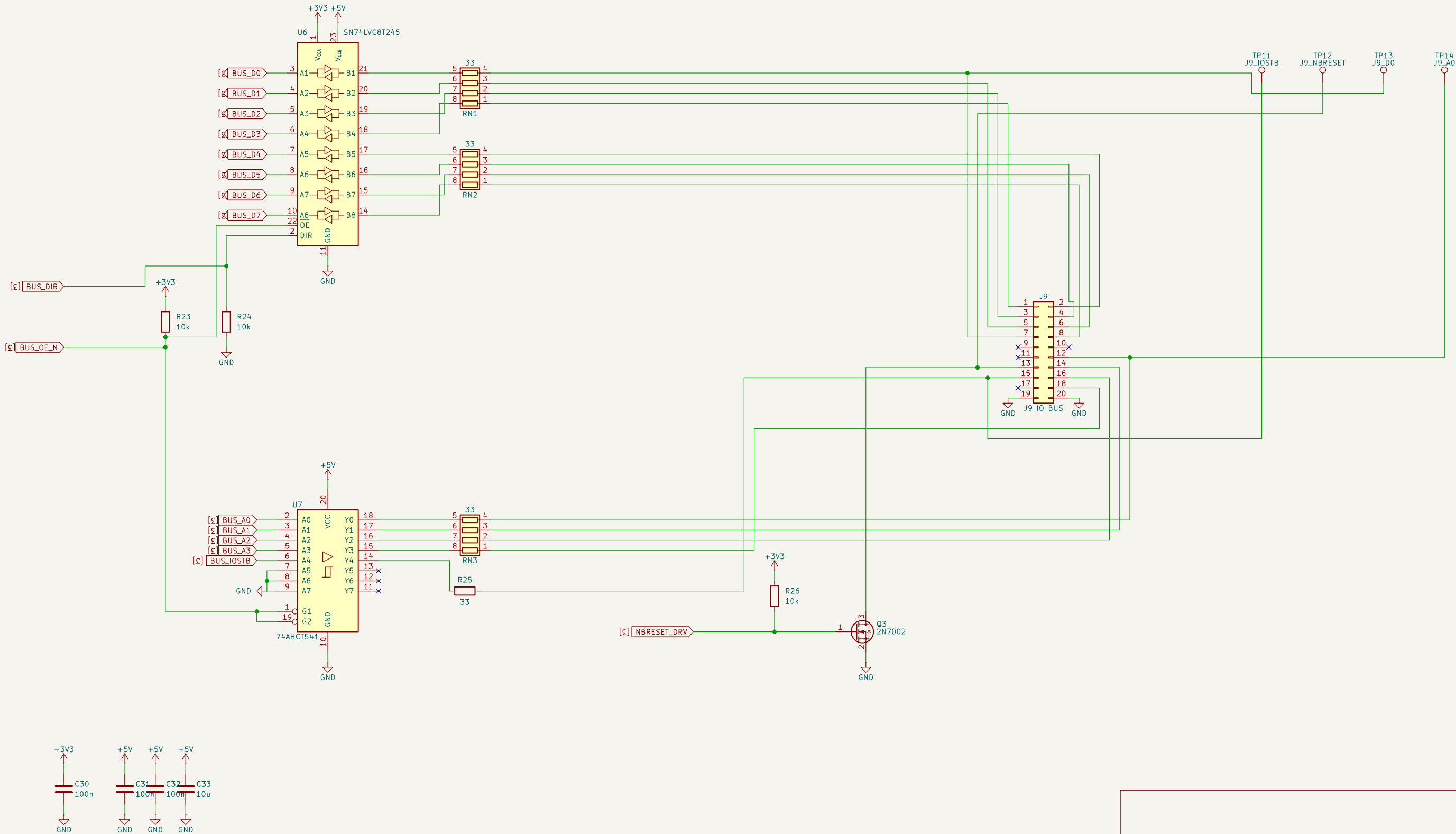
B

C

D

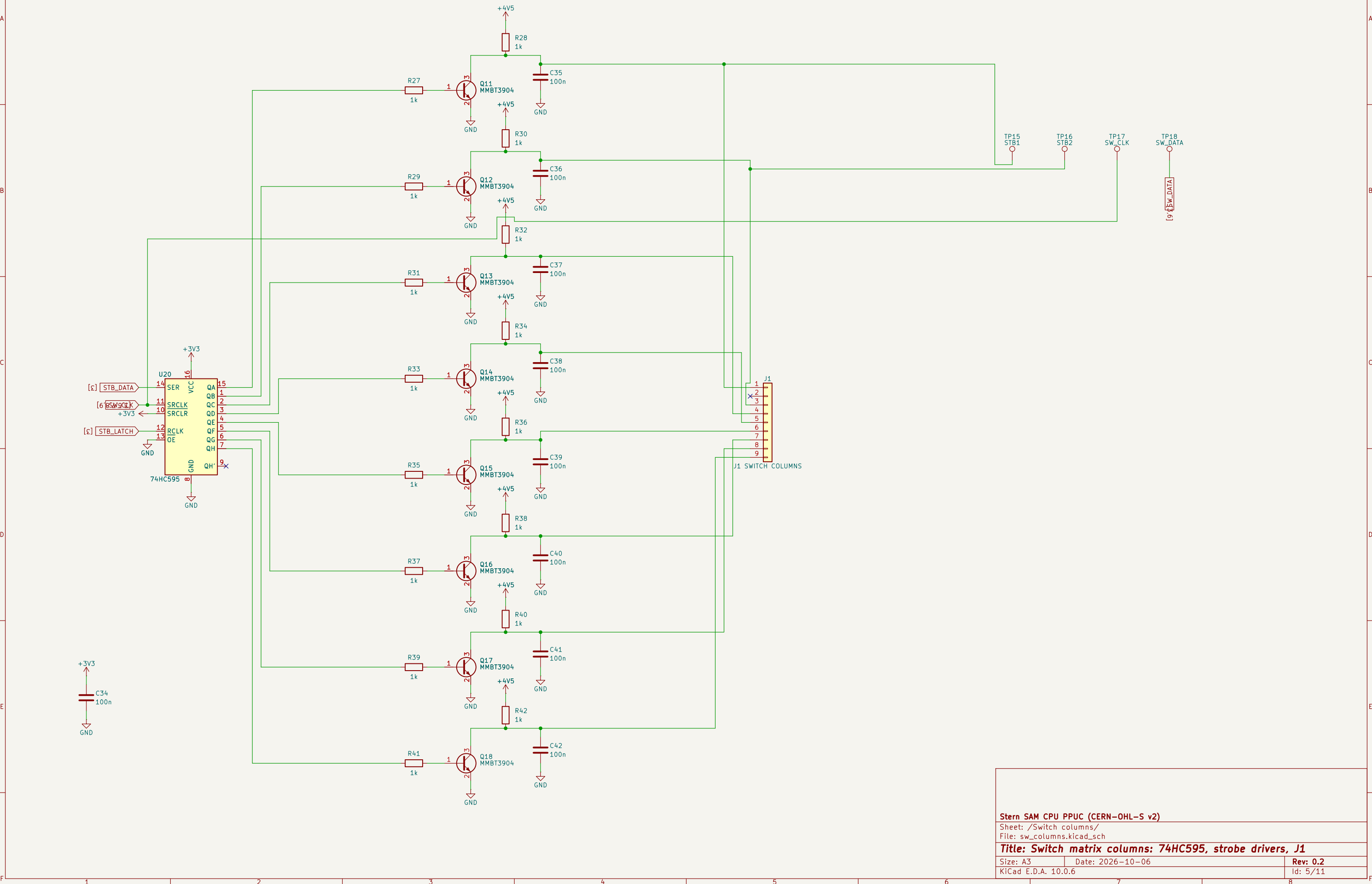
E

F

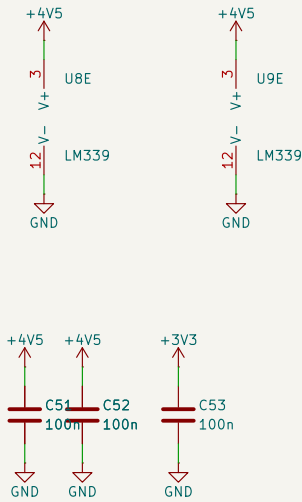
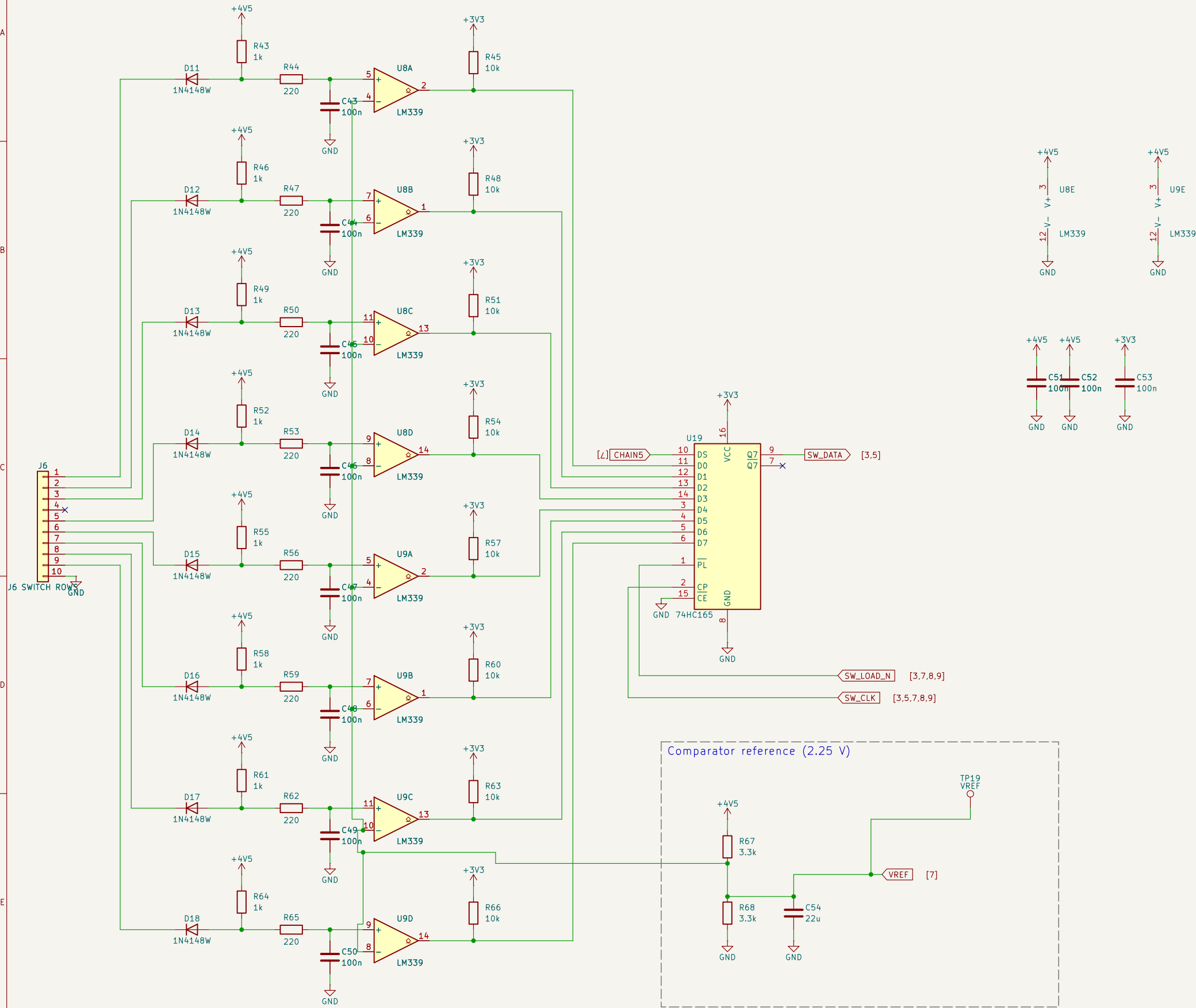


Stern SAM CPU PPUC (CERN-OHL-S v2)		
Sheet: /IO bus/ File: io_bus.kicad_sch		
Title: J9: bus to the IO power driver board (IO board J1)		
Size: A3	Date: 2026-10-06	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 4/11

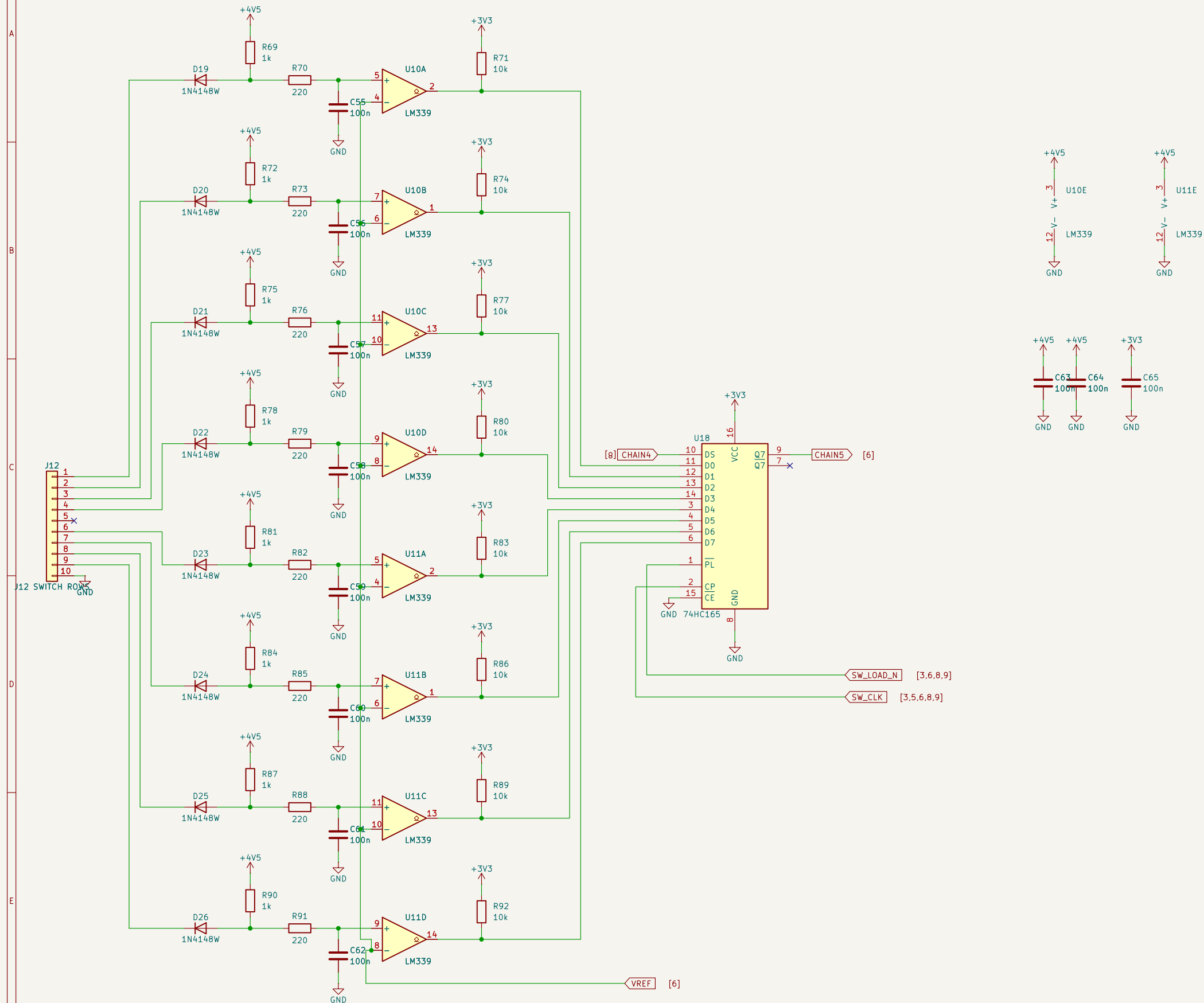
One shift clock for both registers: PIO1 shifts the next strobe pattern into the 595 while the 74HC165 chain shifts in (ARCHITECTURE.md 5.3–5.4).
Strobes: 595 output high → MMBT3904 on → strobe pulled low (active), with the original 1 k pull-up to +4.5 V and 100 nF.
J1 keeps the original pinout (key on pin 2). Strobes 5–8 are driven even where a game does not use them (Q15).



Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.



Per return (ARCHITECTURE.md 5.2-5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing; check it with a meter.



Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Switch rows 9-16/
File: sw_rows_9.kicad_sch

Title: Switch matrix returns 9-16: J12, LM339 front end, 74HC165

Size: A3

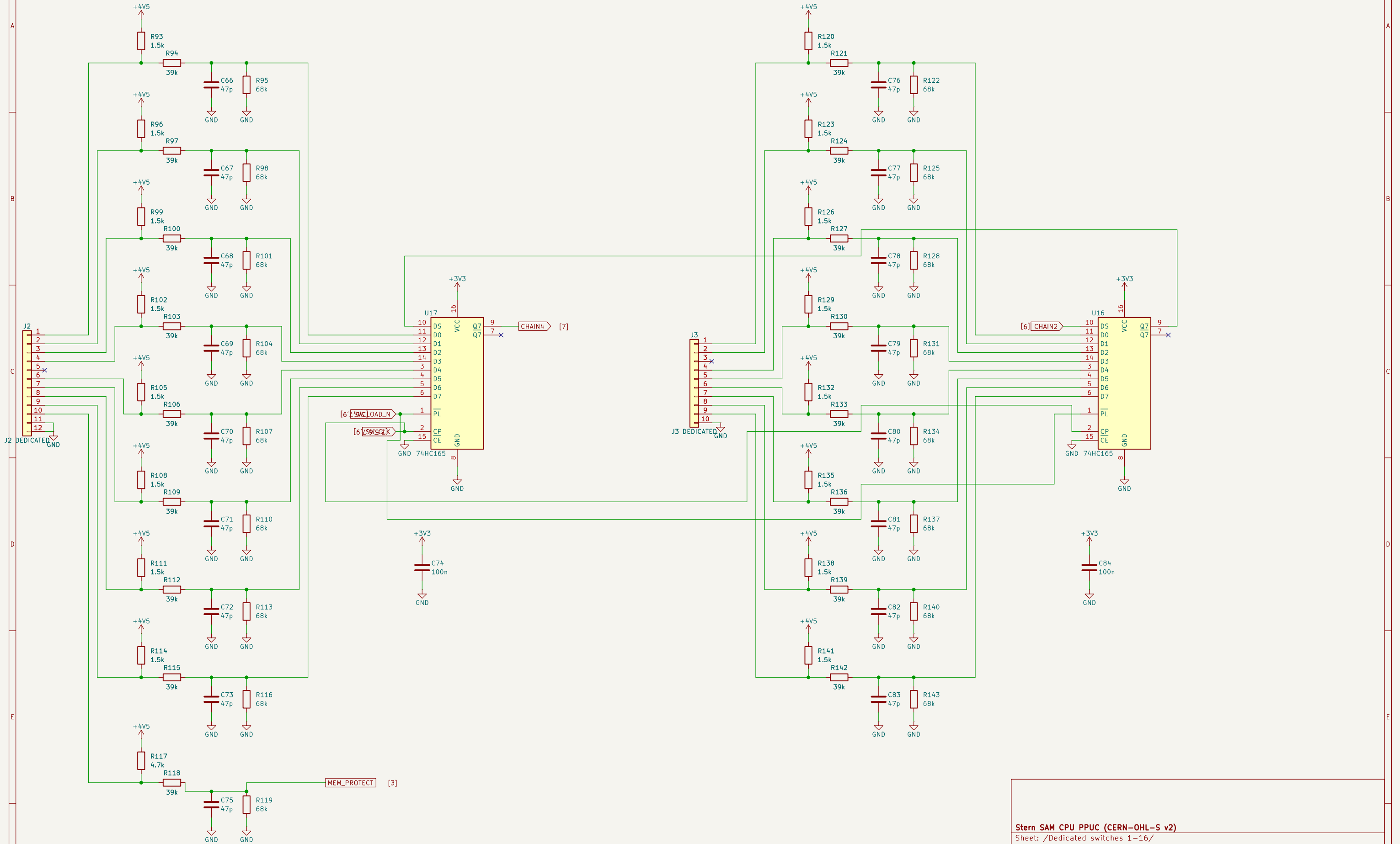
Date: 2026-10-06

Rev: 0.2

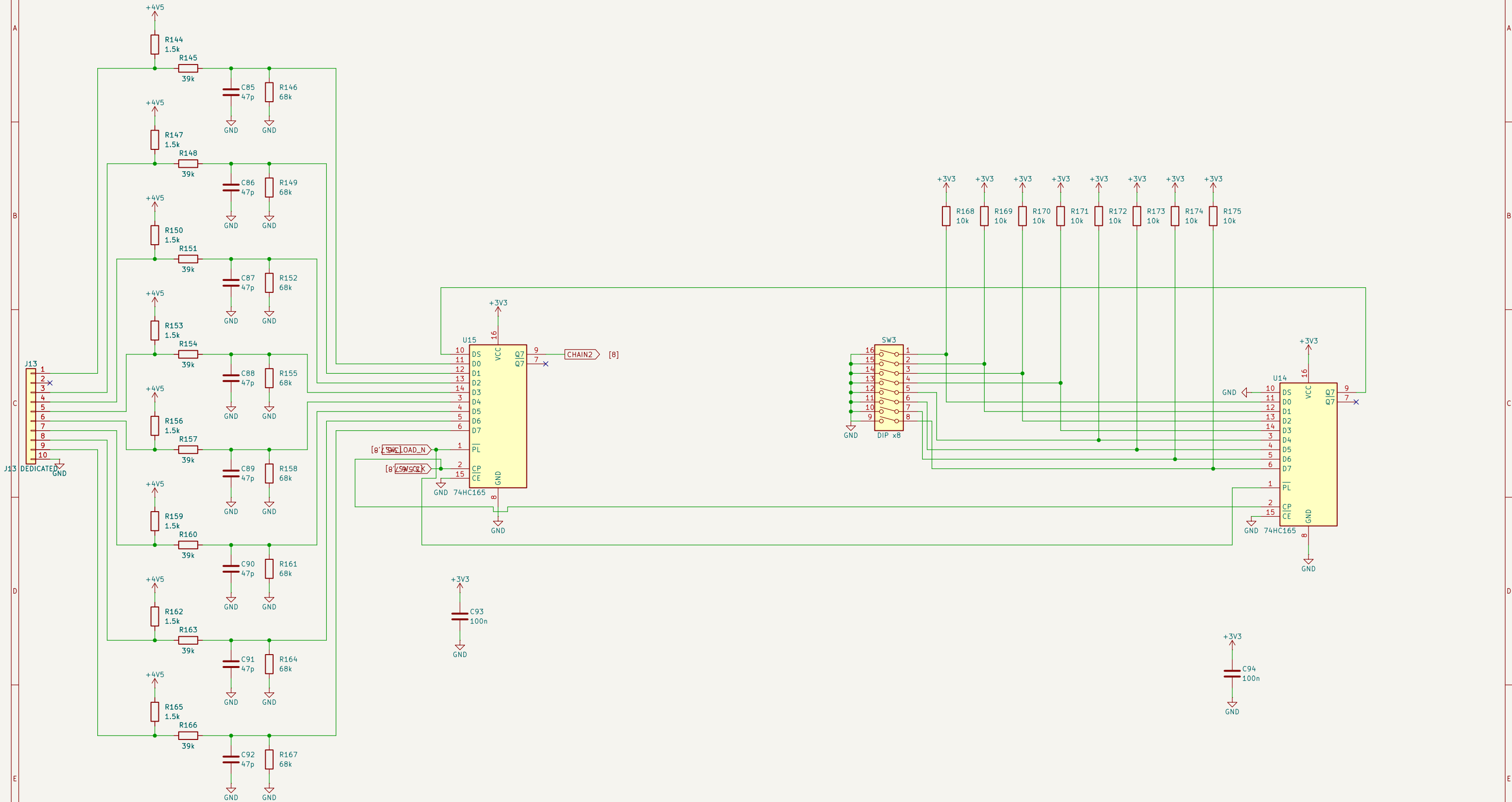
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Rev. 0.2
d: 7/11

Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. Memory protect (J2 pin 10) uses the same filter with the original 4.7 k pull-up and goes straight to RP2354B GPIO31.

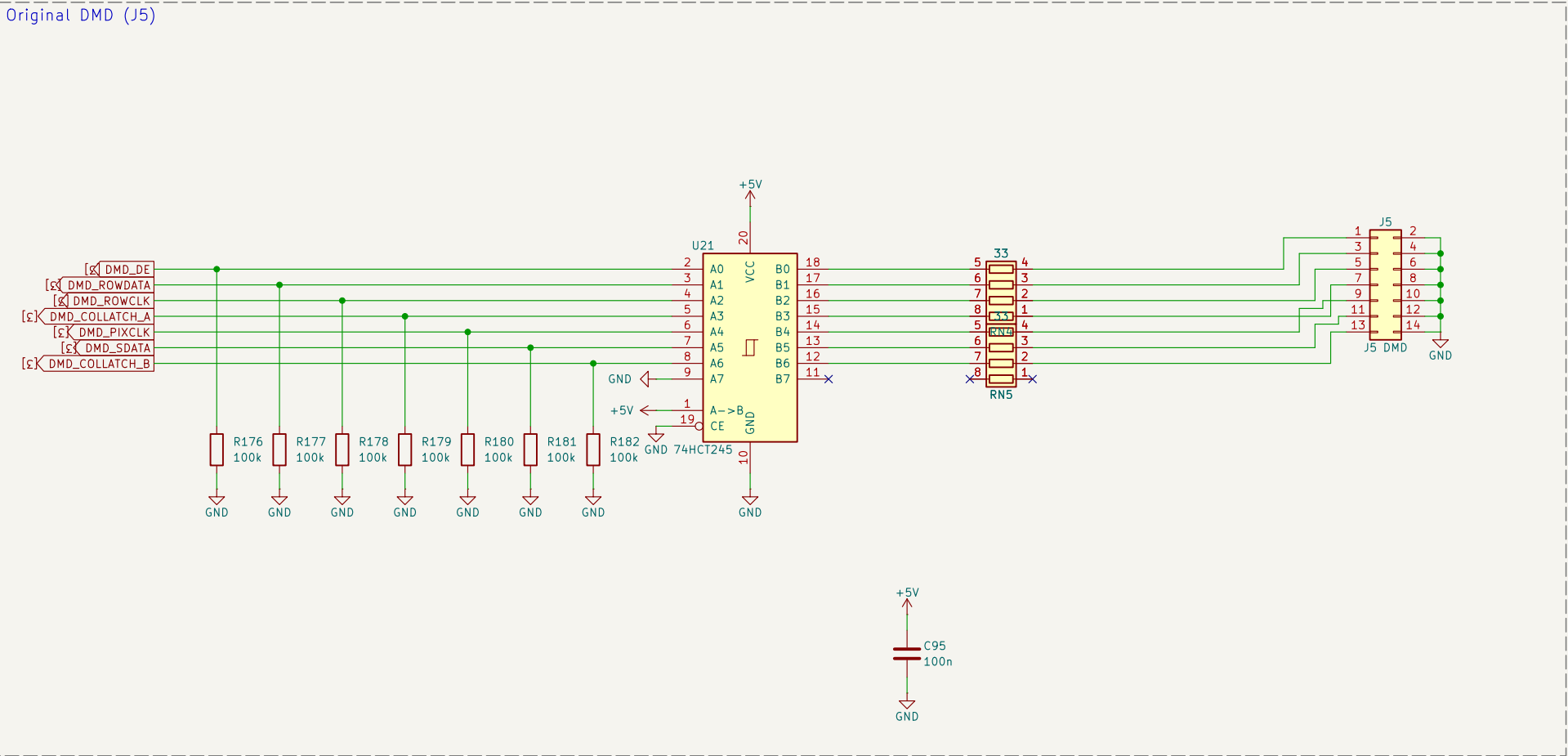


Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. The 8 DIP switches are read through the last 74HC165 of the chain (U14), whose serial input is tied low.

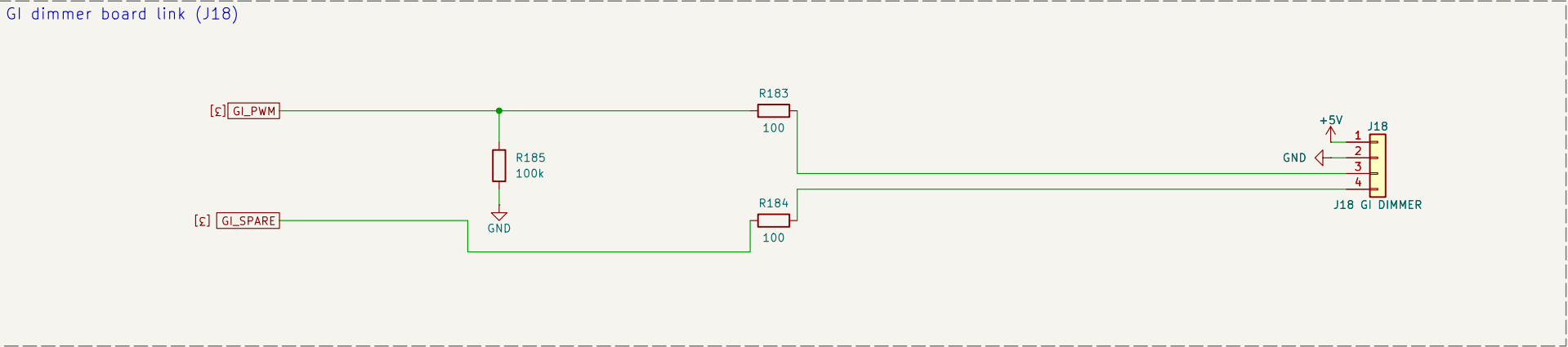


J5 pinout read from the original drawing (ARCHITECTURE.md 8): signals on odd pins, ground on even pins. Confirm with a meter (Q18).
The 100 k pull-downs keep the DMD lines low while the RP2354B boots.
J18 is the default GI dimmer link of ARCHITECTURE.md 10.2: +5 V, GND, GI_PWM, GI_SPARE.

Original DMD (J5)



GI dimmer board link (J18)



Same structure as the original (DAC + two TDA2030A on +-12 V, speakers on J10). Volume stays digital (ROM / PinMAME).
Gain: $22k / 4.7k$ input divider $\times (1 + 22k / 1k) =$ about 4 overall. Starting values, to check on the bench.
J10 keeps the original pinout (left = backbox, right = cabinet woofer by default; the woofer low-pass is done on the Pi).
J22 carries the same two outputs as separate pairs (L+ L- R+ R-) for a stereo backbox harness.

